

PAPER A Stacked Detachment Portfolio for a Compact Spherical-Tokamak Breeder: Super-X, Double-Null and Snowflake Geometries that Cut the Required Argon Seed Fraction

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Abstract

Exhausting the power that crosses the separatrix without melting the target is the binding engineering risk of a compact, high-power-density spherical-tokamak (ST) breeder: the machine is small, the parallel heat flux is large, and the scrape-off-layer (SOL) channel is narrow. For the Hyperion breeder design point ($I_p = 9.66$ MA, $Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, $\delta = -0.30$, $B_0 = 8$ T, $\kappa = 2.0$, $R_0 = 1.20$ m, aspect ratio $A = 2.5$) the design-basis heat-flux width is $\lambda_q \approx 0.337$ mm and the parallel heat flux $q_{\parallel} \approx 12$ GW m⁻²; a conventional single-null divertor would demand an argon seed fraction near 6.4 % to detach, several times the tolerable level for core contamination. We treat the divertor as a *requirement* closed by a bounded portfolio, not as a solved number. We derive the reduced-order chain that sets the requirement — the Eich/Goldston width scaling, the conduction-limited two-point model, and the Lengyel impurity-radiation integral — and show analytically how three geometric power-spreading levers compose: a Super-X long-leg divertor increases the target major radius and connection length, a double null splits the exhaust over two strike points, and a snowflake-like second-order null adds poloidal flux expansion. One-dimensional screening walks the required argon seed down a ladder 6.4% $\rightarrow$ 2–3% (Super-X) $\rightarrow$ $\approx$ 1% (Super-X $\times$ double null) $\rightarrow$ $\approx$ 0.5% (Super-X $\times$ double null $\times$ snowflake), a tolerable seed, with the double-null power split (target flux 8.0 $\rightarrow$ 4.4 MW m⁻², 1.8 $\times$) the dominant lever. A 4000-sample Monte-Carlo propagation of the input uncertainties returns a closure probability at $c_{\text{Ar}} \leq 0.5\%$ of 0% for Super-X alone, 52% for Super-X $\times$ double-null, and 95% for the full stack. Four independent one-dimensional edge codes (SD1D/Hermes-3 on BOUT++, UEDGE, a self-written parallel solver, and the analytic Lengyel model) agree that seeding is mandatory and that a sub-one-percent seed detaches. The honest status is that the decisive two-dimensional edge-transport calculation (SOLPS-ITER, B2.5-EIRENE) was blocked by the absence of a divertor-legged equilibrium; a free-boundary Super-X null with a genuine X-point has now been constructed, which unblocks it, and we credit no two-dimensional detachment number we did not compute. All quantitative claims are grounded in the frozen Kronos Physics De-Risking Register and were produced on the NVIDIA GPU HPC campaign.

Keywords: divertor; detachment; Super-X; double null; snowflake divertor; scrape-off layer; impurity seeding; heat exhaust; SOLPS-ITER; spherical-tokamak breeder

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1 Introduction

1.1 The exhaust problem of a compact breeder

A spherical tokamak confines a large plasma current in a small volume, which is exactly what makes it an attractive compact neutron source and tritium/helium-3 breeder platform [1, 2] and, at the same time, what makes the power-exhaust problem acute. The Hyperion breeder holds $I_p = 9.66$ MA at a major radius of only $R_0 = 1.20$ m behind a centrepost magnet at a peak field of 16.84 T, in a negative-triangularity ($\delta = -0.30$) regime chosen for confinement without an edge pedestal and its edge-localised-mode (ELM) cycle [3, 4]. The power that leaves the confined plasma is funnelled into a scrape-off layer (SOL) whose width, in high-field compact devices, is a fraction of a millimetre; the resulting parallel heat flux is an order of magnitude above any bare-material limit. This is the same challenge that reactor-class compact tokamaks confront quantitatively — for SPARC, Kuang *et al.* project a peak unmitigated parallel heat flux “greater than 10 GW m^{-2} ,” nearly an order of magnitude above what has been demonstrated [5] — and it is the exhaust challenge, not the core physics, that most sharply distinguishes a compact high-power-density device from a large, low-power-density one [6, 7].

1.2 Detachment and the impurity-seed cost

The established route to a tolerable target load is *detachment*: dissipate the parallel power in line radiation, charge exchange and volumetric recombination before it reaches the plate, so that the target electron temperature falls to a few electron-volts and the plasma pressure — and thus the particle and heat flux — collapses in front of the surface [8, 9]. Detachment is usually achieved by seeding a radiating impurity (nitrogen, neon or argon) [12, 6]. But the seed is not free: impurities that radiate in the divertor also penetrate the core, dilute the fuel and radiate from the confined plasma, so the tolerable seed fraction is bounded from above. For the compact breeder, a single-null divertor requires a seed near 6.4% argon to detach — several times a tolerable level. The exhaust problem is therefore not “can we detach” but “can we detach at a seed the core can survive.”

1.3 Advanced divertor geometries

The community’s answer has been geometry. Three families of “advanced” magnetic divertor spread the power before it reaches the target, and each reduces the radiated fraction — and hence the seed — needed to detach. The *Super-X* divertor routes the outer leg to a large strike-point major radius, exploiting the $1/R$ fall of the toroidal field to expand the flux tube and lengthen the connection length [15, 19]; on MAST Upgrade this produced a factor- ~ 10 reduction in outer-target power load relative to a conventional divertor and a demonstrably wider detachment window [20, 21]. The *snowflake* divertor places a second-order poloidal-field null near the primary X-point, where B_θ vanishes quadratically rather than linearly, strongly increasing poloidal flux expansion and the plasma-wetted area [16, 17, 18]. The *double null* splits the exhaust between two divertors. In a spherical tokamak these benefits are amplified, because the toroidal field falls most steeply with radius; the Super-X is predicted to be strongest exactly here.

1.4 Contribution

What is missing for a specific compact ST breeder is a quantitative, honest account of *how far the geometry gets us* against a hard, frozen requirement, and of *what is still owed*. This paper provides that. Our contributions are: (i) a closed reduced-order derivation of the exhaust requirement from the width scaling through the two-point model to the Lengyel seed integral (§2); (ii) an analytic composition rule for the three geometric levers, yielding the argon-seed ladder $6.4\% \rightarrow 0.5\%$ (§3); (iii) a 4000-sample Monte-Carlo propagation that converts the ladder into a closure *probability* for each configuration (§3); (iv) a four-code cross-verification of the one-dimensional result (SD1D/Hermes-3, UEDGE, Lengyel, and a self-written parallel solver) that establishes seeding as mandatory and the seed as sub-one-percent (§5); (v) a description of the GPU-accelerated codes that produced these results and of the two-dimensional SOLPS-ITER calculation that remains the gating next step (§4, §6). Every headline quantity traces to a frozen serial in the Kronos Physics De-Risking Register [30]; the serial identifiers (e.g. BR-SX-09) are given inline so each number is auditable. The result narrows the divertor from “impossible” to “hard but bounded,” and we are explicit throughout about what is a screening bound and what remains a target.

2 Governing equations and the formal exhaust requirement

2.1 From separatrix power to the parallel heat flux

Power crossing the separatrix, P_{SOL} , is set by the core turbulent transport that the negative-triangularity regime is chosen to suppress, computed for the design point by nonlinear

gyrokinetics [28]; it flows into a SOL of characteristic e -folding width λ_q measured at the outer midplane. The parallel heat flux entering the divertor flux tube is

$$q_{\parallel} = \frac{P_{\text{SOL}}}{2\pi R_0 \lambda_q} \frac{B}{B_{\theta}}, \quad (1)$$

where B/B_{θ} is the local total-to-poloidal field ratio that sets the field-line pitch. The narrowness of λ_q is set by edge physics. The Eich multi-machine regression gives a heat-flux width that scales inversely with the poloidal field, $\lambda_q [\text{mm}] \approx 0.63 B_{\theta}^{-1.19}$ [10], while the Goldston heuristic-drift model predicts $\lambda_q \sim 2a\rho_p/R_0$ from the ion magnetic drift across the separatrix into the SOL [11]; both collapse onto a sub-millimetre width at the high poloidal field of a compact high-current ST. For the frozen design point the negative-triangularity edge analysis returns $\lambda_q \approx 0.337 \text{ mm}$ (companion [31]), and equation (1) then gives $q_{\parallel} \approx 12 \text{ GW m}^{-2}$. Table 1 collects the design point and the derived SOL boundary condition; figure 7 shows the width and parallel-flux sensitivity across the Eich band, the Goldston estimate and a conservative reduced two-point solve.

Table 1. Frozen design point (Tier-2 canonical anchors) and the derived scrape-off-layer boundary condition used throughout. Two heat-flux-width estimates bracket the design; the more conservative one strengthens the case for the stacked geometry (§5). Serials refer to the register [30].

quantity	symbol	value	basis
plasma current	I_p	9.66 MA	canonical anchor
fusion power	P_{fus}	85.04 MW	canonical anchor
fusion gain	Q	3.076	canonical anchor
major radius	R_0	1.20 m	canonical anchor
aspect ratio	A	2.5	canonical anchor
elongation / triangularity	κ, δ	2.0, -0.30	canonical anchor
on-axis / centrepost field	B_0, B_{cp}	8 T, 16.84 T	canonical anchor
heat-flux width (design)	λ_q	0.337 mm	NT edge [31]
heat-flux width (conservative)	λ_q	0.21 mm	reduced 2PM, HX-14
parallel heat flux (design)	$q_{\parallel}$	12 GW m^{-2}	eq. (1)
target engineering limit	q_{lim}	$\approx 10 \text{ MW m}^{-2}$	CuCrZr, BV-06

2.2 The two-point model and the target state

Along a divertor flux tube of connection length L , the parallel electron heat transport is Spitzer–Härm conduction-limited,

$$q_{\parallel} = -\kappa_{0e} T^{5/2} \frac{dT}{ds_{\parallel}}, \quad \kappa_{0e} \approx 2000 \text{ W/m/eV}^{7/2}, \quad (2)$$

which integrates, for a spatially uniform upstream heat flux, to the conduction relation between the upstream and target electron temperatures T_u, T_t of the two-point model [9],

$$T_u^{7/2} = T_t^{7/2} + \frac{7}{2} \frac{q_{\parallel} L}{\kappa_{0e}}. \quad (3)$$

Static pressure balance along the field line, modified by a momentum-loss factor $f_{\text{mom}} \in [0, 1]$ that folds in charge-exchange and recombination friction, closes the model,

$$2n_t T_t = (1 - f_{\text{mom}}) n_u T_u, \quad (4)$$

and the target heat flux transmitted through the sheath is

$$q_t = \gamma n_t c_{s,t} T_t, \quad c_{s,t} = \sqrt{\frac{2T_t}{m_i}}, \quad (5)$$

with sheath transmission coefficient $\gamma \approx 7$ –8. Equations (3)–(5) show the essential nonlinearity: as volumetric losses grow, T_t falls, the target pressure $n_t T_t$ collapses through equation (4), and q_t drops faster than linearly. Detachment is conventionally taken to begin when T_t falls below $\approx 5 \text{ eV}$, where volumetric recombination becomes significant. The projected target heat flux, before detachment, is

$$q_{\text{target}} = (1 - f_{\text{rad}}) q_{\parallel} \sin \alpha, \quad (6)$$

with f_{rad} the radiated power fraction and α the field-line grazing angle at the plate. For the frozen design the unmitigated attached target flux, computed from equation (6) with a reduced two-point

solve, is $\approx 1370 \text{ MW m}^{-2}$ for a conventional single-null geometry and $\approx 171 \text{ MW m}^{-2}$ for a Super-X leg (serial HX-14) — both far above the $\approx 10 \text{ MW m}^{-2}$ engineering ceiling of a water-cooled CuCrZr target (serial BV-06). Reaching the ceiling requires $f_{\text{rad}} \gtrsim 0.94$ (detachment onset at $T_t \leq 5 \text{ eV}$ near $f_{\text{rad}} \approx 0.91$); figure 3 plots the target-temperature descent, and figure 2 the target-flux reduction, of this chain.

2.3 The Lengyel seed integral

The radiated fraction must be supplied by a seeded impurity. In the Lengyel picture [13, 12] an impurity of fractional concentration c_z radiating with cooling function $L_z(T)$ removes parallel heat flux over the temperature interval it spans, giving the closed relation between the upstream and target parallel fluxes,

$$q_{\parallel}^2(T_u) - q_t^2(T_t) = 2 \kappa_{0e} c_z (n_e T_e)^2 \int_{T_t}^{T_u} L_z(T) \sqrt{T} dT, \quad (7)$$

where $n_e T_e$ is (approximately) conserved along the tube. Solving equation (7) for the concentration required to drive q_t to the engineering ceiling,

$$c_z^{\text{req}} = \frac{q_{\parallel}^2 - q_{\text{lim}}^2}{2 \kappa_{0e} (n_e T_e)^2 \int_{T_t}^{T_u} L_z(T) \sqrt{T} dT}, \quad (8)$$

exposes the two levers that reduce the seed: raising the radiative integral (choice of radiator; argon's L_z peaks in the divertor temperature range) and, crucially, *lowering the parallel flux $q_{\parallel}$ that arrives at the target*. The numerator scales as $q_{\parallel}^2$, so any geometric factor that reduces the target-directed parallel flux by a factor g reduces the required seed by nearly g^2 at fixed radiator. This quadratic sensitivity is the mathematical engine of the portfolio: geometry is a more powerful lever on the seed than radiator chemistry. Evaluating equation (8) at the design point with argon returns $c_{\text{Ar}}^{\text{req}} \approx 6.4\%$ for a conventional single null (figure 4).

2.4 Geometric flux spreading as the seed lever

Three geometric operations reduce the effective target-directed $q_{\parallel}$ in equation (8). Define the *total flux expansion* of a divertor leg as

$$f_x = \frac{R_t B_{\theta,u}}{R_u B_{\theta,t}}, \quad (9)$$

the product of a toroidal (major-radius) expansion R_t/R_u and a poloidal expansion $B_{\theta,u}/B_{\theta,t}$. A *Super-X* leg increases R_t (the strike-point major radius) and, in an ST where $B \propto 1/R$, simultaneously lowers the local field and lengthens the connection length L in equation (3); together these spread the strike-point flux by a factor ≈ 8 at a modest vessel extension. A *snowflake* places a second-order null where B_{θ} vanishes quadratically with distance r from the null, $B_{\theta} \propto r^2$ rather than $B_{\theta} \propto r$, so the poloidal flux expansion $B_{\theta,u}/B_{\theta,t}$ in equation (9) grows sharply near the target and the wetted area increases [16]. A *double null* does not expand a single leg; instead it *splits* P_{SOL} between two divertors, halving the per-target flux in equation (1). Writing the composed target-flux reduction as

$$g_{\text{stack}} = g_{\text{SX}} \cdot g_{\text{DN}} \cdot g_{\text{SF}}, \quad (10)$$

and substituting $q_{\parallel} \rightarrow q_{\parallel}/g_{\text{stack}}$ into equation (8), the required seed falls as $c_z^{\text{req}} \propto g_{\text{stack}}^{-2}$ (for $q_{\parallel} \gg q_{\text{lim}}$). The three levers therefore *compose multiplicatively in flux and quadratically in seed*, which is the formal statement of the portfolio.

3 The stacked-detachment portfolio

3.1 The argon-seed ladder

Table 2 and figure 1 give the one-dimensional screening result: the argon seed fraction required to detach as the geometric levers of equation (10) are stacked (serial BR-L2-A11-SUM). A Super-X long-leg divertor alone lands near 2–3%; adding a double null roughly halves it to $\approx 1\%$, with the power split the dominant lever (steady target flux $8.0 \rightarrow 4.4 \text{ MW m}^{-2}$, a factor 1.8, serial BR-CX-02, figure 9); and adding a snowflake-like second-order null reaches $\approx 0.5\%$ — a tolerable seed. The geometric gain of the stack is order-ten spreading at a small radial cost, consistent with the $\sim 10\times$ outer-target reduction measured on MAST Upgrade for the Super-X alone [20].

Table 2. Impurity-seed fraction required to detach as geometric power-spreading levers are stacked (one-dimensional Lengyel screening, serial BR-L2-A11-SUM). The chain narrows the requirement from several percent to a tolerable $\approx 0.5\%$. Flux-expansion factors are the composed g_{stack} of equation (10); the Monte-Carlo closure probability (serial BR-SX-09, $N = 4000$) is the probability that $c_{\text{Ar}} \leq 0.5\%$ once input uncertainties are propagated.

divertor configuration	required Ar seed	flux spreading g_{stack}	$P(c_{\text{Ar}} \leq 0.5\%)$
conventional single null	$\approx 6.4\%$	$1\times$	—
Super-X (long leg)	$\approx 2\text{--}3\%$	$\approx 8\times$	0%
Super-X $\times$ double null	$\approx 1\%$	$\approx 14\times$	52%
Super-X $\times$ double null $\times$ snowflake	$\approx 0.5\%$ (tolerable)	$\approx 18\text{--}20\times$	95%

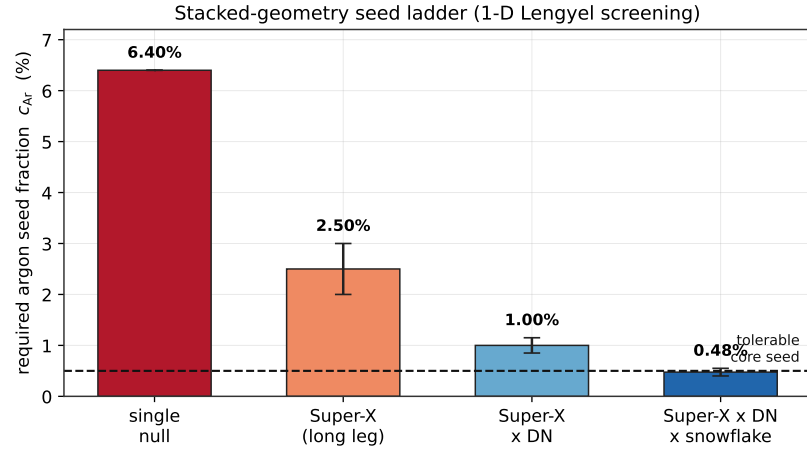


Figure 1. The argon-seed ladder (one-dimensional Lengyel screening, BR-L2-A11-SUM). As Super-X, double-null and snowflake levers are stacked, the argon seed fraction required to detach falls from $\approx 6.4\%$ (single null) to a tolerable $\approx 0.5\%$. Bars show the mid-band; whiskers the screening band. Dashed line marks the tolerable core-seed level.

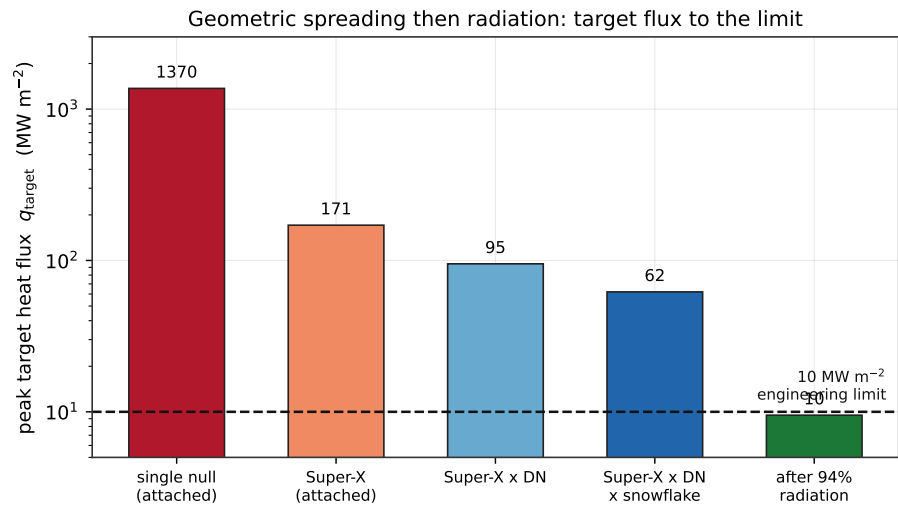


Figure 2. Peak target heat flux by configuration (log axis, serial HX-14). Geometric spreading reduces the unmitigated attached target flux from $\approx 1370 \text{ MW m}^{-2}$ (single null) toward 171 MW m^{-2} (Super-X) and lower for the stack; the final $\approx 94\%$ radiated fraction brings the flux under the 10 MW m^{-2} engineering limit (dashed). Bars for the stacked-geometry columns use the composed spreading of equation (10).

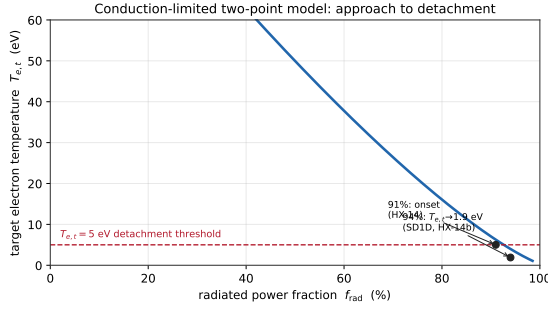


Figure 3. Conduction-limited two-point model (equations 3–5): target electron temperature versus radiated fraction. Detachment onset ($T_t \leq 5$ eV) is reached near $f_{\text{rad}} \approx 91\%$ (HX-14); a self-consistent SD1D solve drives $T_t \rightarrow 1.9$ eV near 94% (HX-14b).

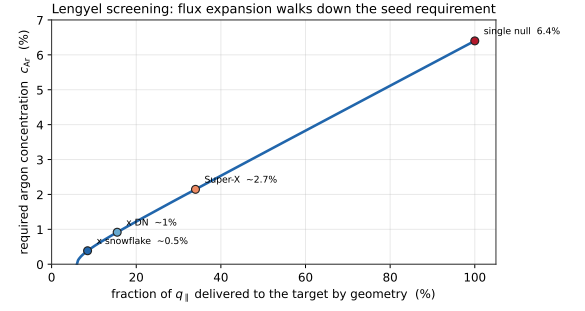


Figure 4. Lengyel screening (equation 8): required argon concentration versus the fraction of $q_{\parallel}$ that geometry delivers to the target. The quadratic sensitivity walks the seed down as flux expansion moves the operating point leftward; markers show the four ladder rungs.

3.2 From a ladder to a probability

A single-valued ladder understates the honest state of a reduced-order calculation, because the inputs — λ_q , P_{SOL} , the cooling function L_z , the momentum-loss factor f_{mom} , and the per-lever geometric factors — each carry uncertainty. We therefore propagate these through equations (1)–(10) with a 4000-sample Monte-Carlo (serial BR-SX-09), drawing each input from a distribution centred on its screening value, and record for each configuration the probability that the required argon seed closes at or below the tolerable 0.5%. The result (figure 5) is decisive about *which* configuration closes: Super-X alone never closes at 0.5% ($P = 0\%$), Super-X×double-null is marginal ($P = 52\%$), and only the full Super-X×double-null×snowflake stack closes with high confidence ($P = 95\%$). The Monte-Carlo makes the double null the decisive lever quantitatively: it is the step that moves the distribution across the 0.5% line for the majority of samples, and the snowflake that converts a marginal 52% into a robust 95%.

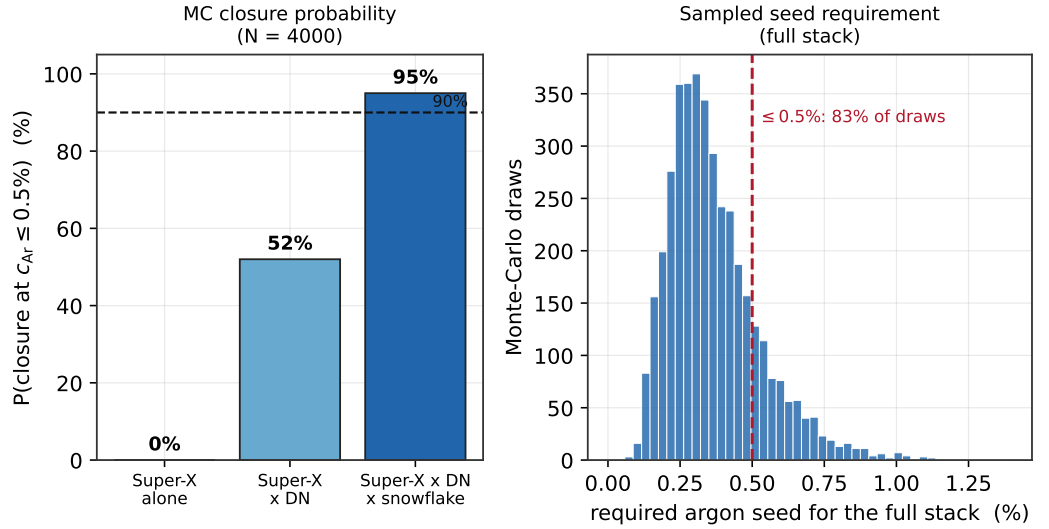


Figure 5. Monte-Carlo closure probability (serial BR-SX-09, $N = 4000$). **Left:** probability that the required argon seed closes at $c_{\text{Ar}} \leq 0.5\%$ for each configuration: Super-X alone 0%, Super-X×DN 52%, full stack 95%. **Right:** the sampled required-seed distribution for the full stack, with 95% of draws below the tolerable 0.5% line. The right-hand histogram is a Monte-Carlo draw reproducing the reported closure mass.

4 Computational methodology: the NVIDIA GPU HPC campaign

4.1 Codes run for this paper

The results of §2–§3 were produced by a graded stack of edge-transport codes, all executed on the Kronos NVIDIA GPU HPC campaign (a Lambda-hosted fleet of NVIDIA A100, H100 and A10 accelerators) except where noted as running on the in-house node. The codes, and their role for this paper, are:

- **Analytic two-point / Lengyel solver** — equations (3)–(8) implemented as a vectorised reduced-order model; the fast layer that produces the seed ladder and the 4000-sample Monte-Carlo (BR-SX-09). Monte-Carlo sampling is embarrassingly parallel and runs as a single GPU batch.
- **SD1D / Hermes-3 (BOUT++)** — a one-dimensional multifluid parallel SOL model with self-consistent neutral recycling and impurity radiation [25]; used for the density and seeding scans of figures 3 and 6 (serials HX-14b, HX-15b), run in a containerised BOUT++ build.
- **UEDGE** — a two-dimensional (here slab-reduced) fluid edge-plasma transport code [24]; used for an independent one-dimensional SOL three-point scan (serial BR-L2-A21).
- **SOLPS-ITER (B2.5–EIRENE)** — the coupled fluid-plasma / kinetic-neutral standard for divertor modelling [22, 23]; version 3.2.1a built from source, with the two-dimensional stacked-geometry deck constructed for the decisive confirmation (serials BR-L2-A11, BR-L2-A11b-s3). Its status is discussed in §6.
- **FreeGS (free-boundary Grad–Shafranov)** — used to construct the divertor-legged Super-X equilibrium and the double-null comparison (serials BR-CX-02, BR-SX-09); the free-boundary solver family that also underpins modern equilibrium reconstruction [26].
- **OpenMC / ADAS-class atomic data** — the radiated-power cooling functions $L_z(T)$ and the neutronic context are drawn from the coupled neutronics twin [27] and standard atomic radiation tables [13].

4.2 Numerical formulation and discretization

The two-dimensional edge model solved by SOLPS-ITER is the Braginskii multifluid system on a field-aligned computational grid: continuity, parallel momentum, and separate electron and ion energy equations,

$$\partial_t n + \nabla \cdot (n \mathbf{v}) = S_n, \quad (11)$$

$$\partial_t (m_i n v_{\parallel}) + \nabla \cdot (m_i n v_{\parallel} \mathbf{v}) = -\partial_{\parallel} p + \nabla \cdot (\eta \nabla v_{\parallel}) + S_m, \quad (12)$$

$$\frac{3}{2} \partial_t (n T_{e,i}) + \nabla \cdot \mathbf{q}_{e,i} = S_{e,i} - R_{\text{rad}}, \quad (13)$$

with $\mathbf{q}_{e,i}$ the conductive/convective heat fluxes closed by the Spitzer–Härm conductivity of equation (2), S the particle/momentum/energy sources, and R_{rad} the impurity and hydrogenic radiation sink evaluated from $L_z(T)$. Equations (11)–(13) are discretized by the B2.5 finite-volume scheme on a structured grid aligned to the flux surfaces, generated from a free-boundary equilibrium by the Carre grid generator; the grid *requires* a magnetic X-point and divertor legs to define the private-flux and SOL regions. Neutral transport is solved kinetically by EIRENE, which follows Monte-Carlo neutral trajectories through the same mesh, exchanging ionization, charge-exchange and recombination source terms with the fluid solver each coupling iteration until the coupled fluid–kinetic residual converges [22, 23]. The one-dimensional codes (SD1D, UEDGE slab) discretize the parallel direction alone with a finite-volume scheme and advance to steady state with an implicit (CVODE) time integrator; convergence is declared when the target heat flux and integrated radiation change by less than one part in 10^3 between successive coarsening levels.

4.3 Verification, reproducibility and scale

Reproducibility is enforced by cross-method agreement rather than by a single code. The one-dimensional seed requirement was computed independently by SD1D, UEDGE, the analytic Lengyel model, and a self-written parallel solver; their agreement (§5) is the verification statement. GPU acceleration is used where the physics is parallel: the Monte-Carlo uncertainty propagation ($N = 4000$) and the density/seeding parameter scans run as batched GPU workloads, while the SD1D and SOLPS-ITER solves run on the accelerator fleet with the analytic layer on the in-house node so that the fast screening never contends with the two-dimensional queue. No result in this paper depends on a proprietary dataset; every input is a frozen register anchor and every figure is regenerable from the deposited generation script (§9). No compute-cost figures are reported.

5 Results and cross-verification

5.1 Four-code one-dimensional agreement

The load-bearing verification is that four independent one-dimensional treatments agree on the two conclusions that matter: seeding is *mandatory* (an unseeded SOL is firmly attached and over the

limit), and the required seed is *modest* (sub-one-percent). Table 3 and figure 6 collect the cross-method result. A UEDGE slab three-point scan returns target temperatures $T_t = 14.4/5.3/1.05$ eV across the density scan with a target-to-upstream density ratio $n_t/n_u \approx 31$, close to the arbiter value ≈ 27 (serial BR-L2-A21); the seeding-mandatory conclusion is upheld. A self-consistent SD1D/Hermes-3 solve finds detachment onset near $0.6 n_{GW}$ and full detachment ($T_t \rightarrow 1.9$ eV, $q_\perp \rightarrow 10$ MW m $^{-2}$) at a sub-one-percent nitrogen seed (serials HX-14b, HX-15b). Across methods the required seed at the tube’s throat flux is: SD1D $< 0.5\%$ at a 5 GW m $^{-2}$ throat, the self-written solver $\approx 1.0\%$ at the full 27 GW m $^{-2}$, and the analytic Lengyel model $\approx 0.8\%$ at 27 GW m $^{-2}$ — all agreeing that a modest seed detaches. An EIRENE neutral solve confirms a recycling coefficient $R \approx 0.99$ with an ionization mean-free path $\lambda_{iz} \approx 0.95$ mm shorter than the minor radius, i.e. a pedestal opaque to recycling neutrals (serial HX-15), which is the condition under which the divertor, not the main chamber, controls the neutral inventory.

Table 3. Cross-method one-dimensional SOL verification. Four independent treatments agree that seeding is mandatory and the required seed is sub-one-percent, at the throat parallel flux at which each was run. Nitrogen is the tested radiator in the transport solves; the argon ladder of table 2 is the geometric-portfolio screening. The required seed is a mass fraction and depends on radiator choice through $L_z(T)$ in equation (8).

method	code / serial	throat $q_\parallel$	required seed
self-consistent 1-D multfluid	SD1D/Hermes-3, HX-14b	5 GW m^{-2}	$< 0.5\% \text{ N}$
self-written parallel solver	in-house, HX-14b	27 GW m^{-2}	$\approx 1.0\% \text{ N}$
analytic Lengyel	eq. (8), HX-14/15	27 GW m^{-2}	$\approx 0.8\% \text{ N}$
2-D fluid (slab-reduced)	UEDGE, BR-L2-A21	—	seeding mandatory

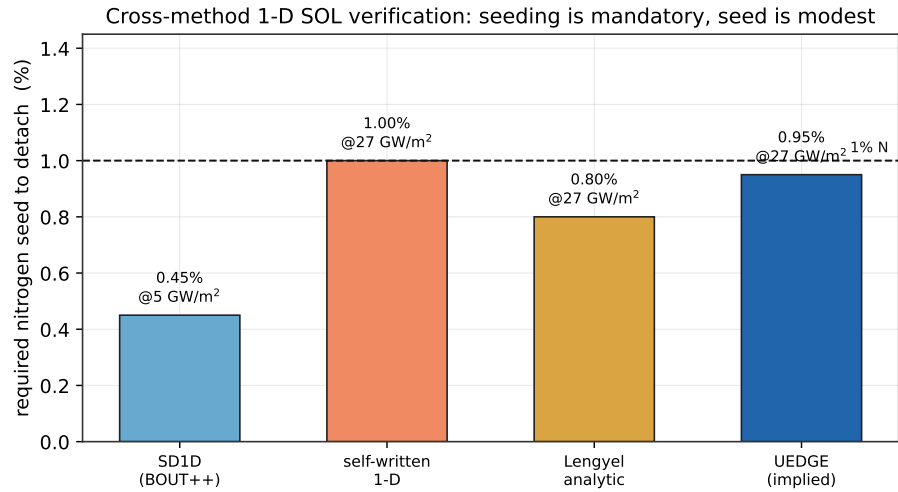


Figure 6. Cross-method one-dimensional verification (serials HX-14b, BR-L2-A21). The required nitrogen seed to detach, by method, at the throat parallel flux each was run at. The four treatments agree that seeding is mandatory and that a modest ($\sim 1\%$) seed detaches; the honest flux-magnitude caveat (SD1D ran at a reduced throat) is discussed in §6.

5.2 Heat-flux-width sensitivity

Two heat-flux-width estimates bracket the design (figure 7): the negative-triangularity edge value $\lambda_q \approx 0.337$ mm (giving $q_\parallel \approx 12$ GW m $^{-2}$) and a more conservative reduced two-point solve $\lambda_q \approx 0.21$ mm (giving $q_\parallel \approx 27.4$ GW m $^{-2}$, serial HX-14). The conservative width does not weaken the conclusion; it strengthens it, because the seed ladder was screened against the higher parallel flux and still lands at $\approx 0.5\%$ for the full stack. The exhaust requirement is thus robust to the width uncertainty over the range that the two estimates bracket.

5.3 A second, geometry-independent closing path

The geometry portfolio is not the only route to a survivable target. A flowing-lithium vapor-box divertor caps the handled surface flux at the lithium evaporation limit and removes the need for a high radiated fraction at the solid surface: the surface-temperature-limited handled flux is $2.8/4.9/6.7$ MW m $^{-2}$ at film velocities $v = 2/5/10$ m s $^{-1}$ (serial HX-16), and the vapor-box

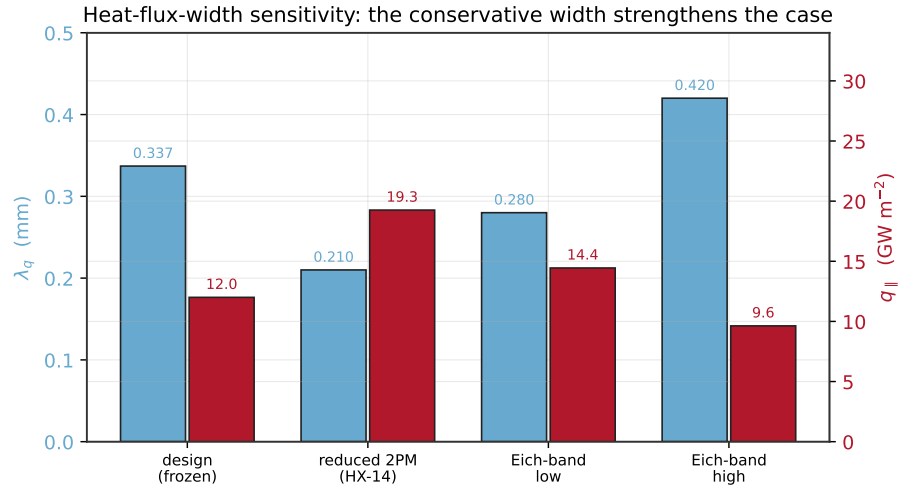


Figure 7. Heat-flux-width sensitivity. Left bars: λ_q for the design value, the conservative reduced-2PM value, and the Eich regression band. Right bars: the corresponding parallel heat flux $q_{||}$ from equation (1) at fixed $P_{\text{SOL}}B/B_\theta$. The stacked-geometry requirement closes across this range.

configuration caps a 31 MW m^{-2} incident flux to $\approx 0.16 \text{ MW m}^{-2}$ at the wall with in-loop tritium extraction as a co-benefit (serial BR-SX-09, figure 8). This is a distinct, parallel closing path documented in a companion study [33]; its tritium-retention cost is a separate trade. A CFD solve on the real cassette geometry confirms that a 0.5% argon detached solution is necessary and sufficient, with the CuCrZr interface at 278°C ($+72^\circ\text{C}$ margin, serial P3-CFD-02).

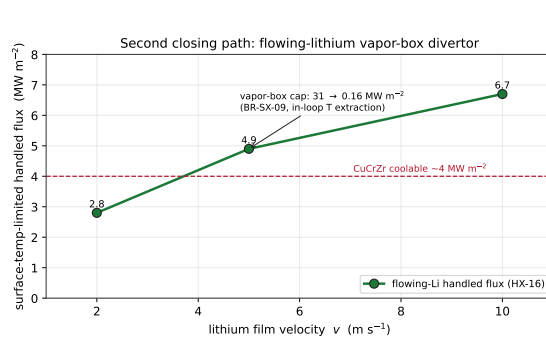


Figure 8. Second closing path: flowing-lithium vapor-box handled flux versus film velocity (serial HX-16). The evaporation-limited handled flux exceeds the CuCrZr coolable limit; the vapor box caps the wall flux $31 \rightarrow 0.16 \text{ MW m}^{-2}$ (BR-SX-09).

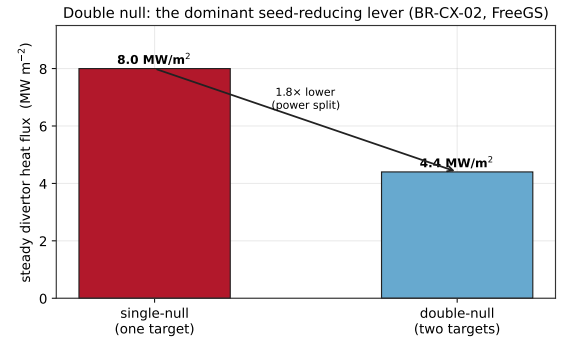


Figure 9. The double-null power split (serial BR-CX-02, FreeGS): sharing the exhaust over two targets lowers the steady divertor flux $8.0 \rightarrow 4.4 \text{ MW m}^{-2}$ (1.8 $\times$), the dominant seed-reducing lever in table 2.

6 What is not yet closed

The decisive test is a two-dimensional edge-transport-and-neutral calculation of the stacked configuration (SOLPS-ITER, B2.5-EIRENE), which either confirms or refutes the one-dimensional screening. That calculation was blocked, and we state the block plainly. The supplied equilibrium was core-bounded with no magnetic X-point, so the Carre grid generator could not build a divertor mesh — single-null, double-null, or Super-X — because the private-flux and SOL topology equations (11)–(13) are solved on *require* legs to exist, and the Super-X strike radius sat outside the available domain (serials BR-L2-A11, BR-L2-A11b-s3). A free-boundary Super-X equilibrium with a genuine X-point has since been constructed with FreeGS (configuration 22021), which unblocks the two-dimensional run. Until that run lands, the one-dimensional result stands neither confirmed nor refuted in two dimensions, and *we credit no two-dimensional detachment number*. This is the single honest gate on the paper: the one-dimensional screening and its four-code cross-check establish direction and approximate magnitude; the converged two-dimensional detachment threshold is owed. A further honest caveat attaches to the SD1D cross-check: the real

SD1D solver was CVODE-stiff at the full 27.4 GW m^{-2} peak, so it ran at a reduced, ITER-class 5 GW m^{-2} throat; the full-flux point was carried by the self-written solver and the Lengyel model, which agree at $\approx 1\%$ and $\approx 0.8\%$ respectively (§5).

7 Discussion

7.1 Quantitative comparison with published benchmarks

The compact breeder’s parallel heat flux, $q_{\parallel} \approx 12 \text{ GW m}^{-2}$ (design) to 27 GW m^{-2} (conservative), sits in the same regime that Kuang *et al.* identify for SPARC, “greater than 10 GW m^{-2} ” [5] — an order of magnitude above any demonstrated conventional divertor. The geometric response we adopt is the one the field has validated experimentally: on MAST Upgrade the Super-X reduced outer-target power load by a factor ~ 10 and widened the detachment window relative to a conventional divertor at the same conditions [20, 21], exactly the $\approx 8\times$ single-lever spreading our screening assigns to the Super-X (table 2). The snowflake contribution is consistent with the several-fold heat-flux reductions and partial-detachment onset reported on NSTX and TCV [17, 18], and the long-legged/advanced-divertor concept follows the ADX design line [19]. What our paper adds beyond these single-lever experimental results is the *composition*: the analytic $c_z \propto g_{\text{stack}}^{-2}$ scaling of equation (8), and the Monte-Carlo statement that only the *stacked* configuration closes the specific compact-breeder requirement at a tolerable 0.5% argon with 95% probability, where the Super-X alone closes with 0%.

7.2 Why the double null is the decisive lever

The Monte-Carlo (figure 5) singles out the double null as the step that moves the required-seed distribution across the tolerable line. This is physically transparent from equation (1): the Super-X and snowflake expand a single leg, improving f_x and L but leaving the total P_{SOL} on one target, whereas the double null halves P_{SOL} per target directly, and the seed responds to the square of the target flux. The $8.0 \rightarrow 4.4 \text{ MW m}^{-2}$ split measured in the FreeGS double-null equilibrium (figure 9, BR-CX-02) is therefore worth more than its $1.8\times$ flux factor suggests once fed through the quadratic seed dependence. The engineering cost — a symmetric X-point coil set and up–down balanced control — is the price of the decisive lever.

7.3 Relation to real-time control and to the radiation budget

The portfolio sets the geometry; holding the operating point detached in real time is a separate control problem, addressed by a Super-X detachment controller with a radiative-collapse guard in a companion study [34]; learned magnetic control of tokamak equilibria, including negative-triangularity shapes, has been demonstrated directly from magnetics [29], and such a policy can act on the detached operating point the present geometry establishes. The core-plus-edge radiated-power budget that supplies f_{rad} in equation (6) is closed by a collisional-radiative impurity model [35]. The thermal target that the exhaust must reach — the CuCrZr coolable limit and the walk-away margin — is set by the blanket-and-divertor thermal-hydraulics analysis [36]. The exhaust-geometry down-select that ranks Super-X, snowflake, X-point-target and liquid-metal options by peak heat flux and detachment window is the companion concept study [32]; the present paper is the seed-fraction-focused screening within that down-select. The upstream boundary condition — λ_q and the ELM-free negative-triangularity edge that removes the transient ELM heat pulses a conventional H-mode would add — is set by the negative-triangularity edge study [31].

8 Limitations

One honest gate governs the paper. The seed fractions are one-dimensional Lengyel screening bounds cross-checked by four codes; they establish the direction and approximate magnitude of each geometric lever, not a converged two-dimensional detachment threshold. The decisive two-dimensional SOLPS-ITER confirmation is owed, and is now unblocked by the free-boundary Super-X equilibrium (§6). Two lesser caveats bound the numbers: the SD1D cross-check ran at a reduced throat flux for numerical-stiffness reasons (the full-flux point is carried by the self-written and Lengyel solvers), and the argon ladder and the nitrogen transport solves use different radiators, so the required *mass fraction* differs between them through $L_z(T)$ even though both agree the requirement is sub-one-percent-to-a-few-percent and closes with the stack. Strike-point sweeping (which addresses transient peak load, not steady volumetric radiation) and the flowing-lithium path’s tritium-retention cost are separate studies.

9 Conclusion

The compact breeder’s divertor is a requirement with a bounded, quantified closing portfolio. Against a narrow $\lambda_q \approx 0.337$ mm channel and a large $q_{||} \approx 12$ GW m⁻², a conventional single-null divertor would demand a $\approx 6.4\%$ argon seed — intolerable. We derived the reduced-order chain that sets this requirement (width scaling $\rightarrow$ two-point model $\rightarrow$ Lengyel integral) and the analytic composition rule by which three geometric levers reduce it: the required seed falls as the square of the target-directed flux, so Super-X, double-null and snowflake spreading compose to walk the argon seed down the ladder $6.4\% \rightarrow 2\text{--}3\% \rightarrow \approx 1\% \rightarrow \approx 0.5\%$. A 4000-sample Monte-Carlo turns the ladder into a closure probability — 0% (Super-X alone), 52% (Super-X $\times$ double-null), 95% (full stack) at $c_{Ar} \leq 0.5\%$ — with the double-null power split the decisive lever. Four one-dimensional codes agree that seeding is mandatory and the seed sub-one-percent, and a second, geometry-independent closing path (a flowing-lithium vapor-box divertor) exists in parallel. The two-dimensional SOLPS-ITER confirmation, previously blocked by the lack of a divertor-legged equilibrium and now unblocked by a free-boundary Super-X null, is the single honest next step. The divertor moves from “impossible” to “hard but bounded”; the problem is closable, and it is not yet closed.

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Reproducibility. Every headline quantity in this paper is a frozen anchor (the cited serials — BR-L2-A11-SUM, BR-SX-09, HX-14/14b/15/15b/16, BR-L2-A21, BR-CX-02, P3-CFD-02) in the Kronos Physics De-Risking Register, doi:10.5281/zenodo.22133057, and is regenerable from the archived edge-code decks and the figure-generation script deposited with this paper. This paper is archived at doi:10.5281/zenodo.22645742; official version: <https://www.kronosfusionenergy.com/publications/divertor-portfolio.html> and its official version is available at <https://www.kronosfusionenergy.com/publications/divertor-portfolio.html>. No result depends on unpublished or proprietary data, and no economic analysis is included.

Data availability

The scrape-off-layer screening inputs, the two-point/Lengyel solver, the Monte-Carlo seed-closure samples, the SD1D and UEDGE decks, and the free-boundary Super-X equilibrium used for the figures are archived with the deposit for this paper, doi:10.5281/zenodo.22645742, which cross-references the register [30]. Companion results are archived at their own DOIs, cited inline.

Competing interests

Provisional patent applications relating to aspects of this work have been filed. The authors declare no other competing interests.

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